Leviathan: Autonomous Ocean Floor Mapping Technology

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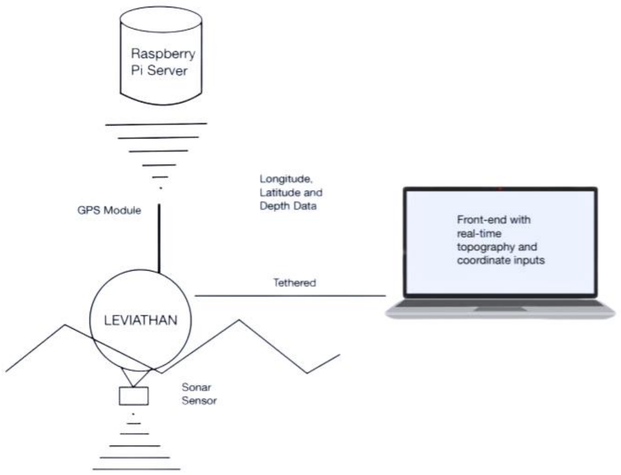
OUR PROJECT

This prototype autonomously maps the ocean floor efficiently and cost-effectively. Leviathan utilizes a repurposed buoy to float on the surface of the water, implementing an ultrasonic sensor and GPS module to collect ocean depth and location data. This data is then relayed to the Raspberry Pi's server and a 3D topographical map is generated using a Python GUI. This tool will provide safer navigation and significant advancements in marine research. Our market ranges from naval operations to marine scientists. This technology will enhance naval operations, marine research, and maritime security by providing a valuable tool for safe navigation, exploration, and scientific understanding of the ocean environment.

python™

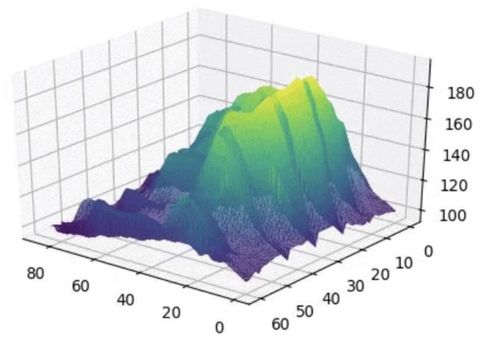
SCALABILITY

Currently, Leviathan can yield real-time depth data for depths up to 6m and can produce a 3D topographical map of the shore of Arbutus Cove during high tide. The future scope of this project aims to yield distance data for higher depths with wireless communication by employing a LoRa module, as well as implementing a small solar module for power independence. These changes would allow Leviathan to be deployed in more diverse environments with minimal maintenance costs.



METHODS

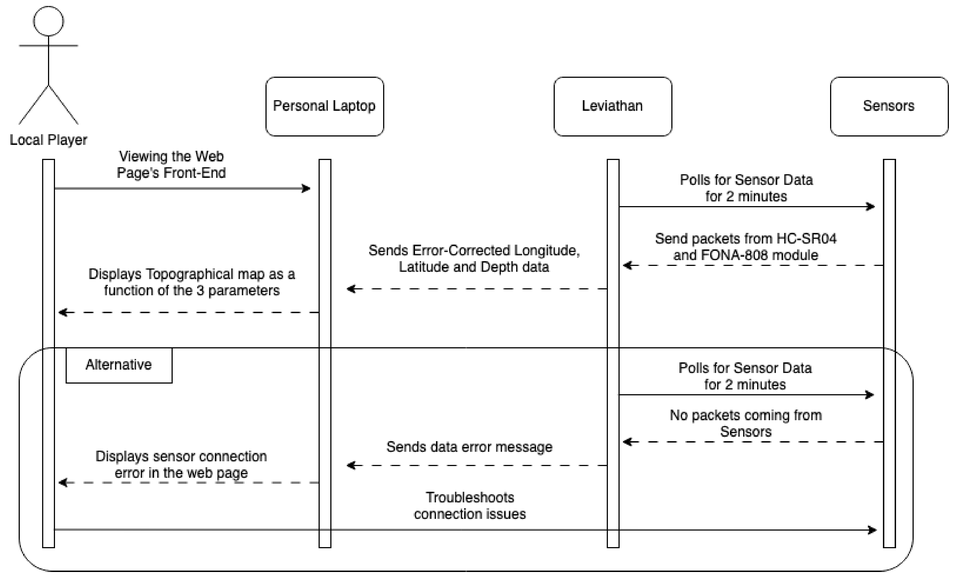
Our prototype employs two types of sensors – a sonar sensor and a GPS. These sensors are essential for generating 3D topographical maps in real time. The maps rely on three variables: longitude, latitude, and ocean floor depth. The data collected by the sensors is parsed through various Python scripts to ensure that it is in the proper format for the mapping software. The 3D topographical map is made with a number of Python libraries. The software is locally hosted with an ethernet cable for ease of use and increased cybersecurity.



ENVIRONMENTAL INITIATIVES

The initial enclosure design was proposed as a 3D printed catamaran style chassis; however, our mentor provided the design solution of repurposing a decommissioned buoy as our enclosure. This promotes eco-friendly initiatives while cutting costs by repurposing ocean garbage into a device that will collect meaningful data and positively impact ocean sciences.





MATERIALS

The main components used in the design of our prototype are as follows:

- Raspberry Pi 4 microcontroller
- Waterproof ultrasonic distance sensor
- Adafruit GPS module

ACKNOWLEDGEMENTS

Our team would like to acknowledge our mentor, Andrew Polanyi, for his input and support as we built this project. We would also like to acknowledge Professor Navneet Popli for setting up this design opportunity and the team at CCIC for their financial contributions to the project.

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